	ADNOC GAS **EPC WORKS** **MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM)** **INSPECTION TEST PLAN**	

DOCUMENT No.: SDOC-1000482-GG-542-1310-A01-SNY-07-007-A4	**Rev.:** A	
DISCIPLINE: Static Equipments	**DOCUMENT TYPE:** Vendor Data	**CLASS:** 1
VDDR CODE: PLN-0001	**VENDOR:** SNY VALVE (YANCHENG) CO., LTD.	
P.O. Nº: 1061008020	**EQUIPMENT:** CAST VALVES CS AND AS < 24"	**UNIT:**

ADNOC GAS

INSPECTION TEST PLAN

AG PROJECT NO.: 1000482
CONTRACTOR PROJECT NO.: 10610

EPC WORKS
MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM)

REVIEW RESPONSE BY CONTRACTOR:

ADNOC -CODE- 2-COMMENTS-TRANSMITTAL NUMBER WF-009623

	A - REJECTED
X	B - REVIEW WITH MAJOR COMMENTS
	C - REVIEW WITH MINOR COMMENTS
	F - REVIEW WITHOUT COMMENTS
	I - FOR INFORMATION
	R - AS BUILT
	M - VOID / CANCELLED

Name:	RIAZ AHAMED
Discipline:	Inspection
Date:	27--07-2024

Under the contractual frame of the PO/Subcontract, CONTRACTOR review comments do not indicate either responsibility or liability for accuracy completeness of this Document, or alter any contractual term and condition before VENDOR, or SUBCONTRACTOR. However, under the contractual frame of the AGREEMENT, CONTRACTOR remains liable before COMPANY.

These are preliminary comments , Vendor to note drawings are not submitted and fully approved from Engineering ,Submit next revision with CRS for each points

Provide individual sections for each type of valve

Include GAD number & Datasheet number

Inspection and Test Plan

INSPECTION AND TEST PLAN

Inspection Class: CRITICALITY Rating:CR1&CR2

MANUFACTURER NAME : SNY VALVE (YANCHENG) CO.,LTD.	**PROJECT / ORDER NO :** EPC WORKS MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM) 1001008020	**ITP NO. :** 1001068020-ITP-R0
DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)	**SUB-SUPPLIER :**	**SUB-ORDER NO. :**

AR Approval Required
A1-A2 Actual Inspection
W1-W2 Witness Inspection
S Operation Surveillance
R1-R2 Review of Documentation
T/C Test Certificate Required
H Hold Point
M Monitoring
1=100%, 2=10%

VENDOR TPI FOR 3.2 CERTIFICATION

CONTRACTOR TPI

COMPANY TPI

ACTION	QUALITY RELATED ACTIVITY	REFERENCE DOCUMENT	CHARACTERISTICS TO BE VERIFIED	ACCEPTANCE CRITERIA	VERIFYING DOCUMENT (*Note 1/2)	INTERVENTION POINTS				
						SUB-CONT.	SNY	~~TPI ADNOC~~	CONTRACTOR / QC	COMPANY
1.0	**DOCUMENT REVIEW**									
1.1	Review and approval of Valve General Assembly Drawing & Inspection and Test Plan	1000482-ADD-SP-09-003,Rev.0 API STD 594 9th ED (2022) API STD 600 14th ED (2021) API STD 623 2nd ED (2021) API SPEC 6D 25th ED (2021) ASME B16.34-2020 SPP-1000482-GG-500-13-45-006,Rev.1 SPP-1000482-GG-500-13-45-007,Rev.0 SPP-1000482-GG-500-13-45-001,Rev.0 NACE MR0175/ISO 15156-2015	Quality document, procedure, test method, inspection/test frequency and acceptance criteria shall be clarified and included.	1000482-ADD-SP-09-003,Rev.0 API STD 594 9th ED (2022) API STD 600 14th ED (2021) API STD 623 2nd ED (2021) API SPEC 6D 25th ED (2021) ASME B16.34-2020 SPP-1000482-GG-500-13-45-006,Rev.1 SPP-1000482-GG-500-13-45-007,Rev.0 SPP-1000482-GG-500-13-45-001,Rev.0 NACE MR0175/ISO 15156-2015	General Arrangement Drawings Inspection and Test Plan approval status	/	H	/	AR	AR
1.2	Review and approval of QA/QC Procedures as per VDRL	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 AGES-GL-13-002,Rev.1 1000482-ADD-SP-07-004,Rev.C 1000482-ADD-SP-07-003,Rev.C DGS 6710 001 Rev.2 NACE MR0175/ISO 15156-2015	QA/QC Procedures: Welding/CRA Overlay Welding Procedure/WPS/PQR NDE Procedures – PT/MT Hydro Test Procedure Fugitive Emission Test Procedure Production Control Procedure FAT Procedure Storage, Packing and Preservation Procedure Painting Procedure	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 AGES-GL-13-002,Rev.1 1000482-ADD-SP-07-004,Rev.C 1000482-ADD-SP-07-003,Rev.C DGS 6710 001 Rev.2 NACE MR0175/ISO 15156-2015	QA/QC Procedures approval status	/	H	/	AR	AR
1.3	Pre-inspection Meeting (PIM)	As per purchase order	Pre-inspection Meeting	PIM agenda	Minutes of Meeting	/	H	/	H	H
2.0	BEFORE MANUFACTURING: MATERIAL TEST CERTIFICATE REVIEW									
2.1	Review and approval of personnel Qualifications & certificates	DGS 0000 003 Rev.2 1000482-ADD-6300-001,Rev.0	NDE Qualification Certificates as per ASNT-TC-1A , Welders qualification certificates	DGS 0000 003 Rev.2 1000482-ADD-6300-001,Rev.0	Qualification Certificates of NDE Personnel/Inspection Personnel	/	H	/	R	R
2.2	Reviewing Raw Material Test Certificates of Components, Heat treatment Records & charts, Hardness test reports, IGC reports, Impact test reports, PMI reports Parts：body, bonnet, Wedge, disc, stem, seat ring	ASTM A105-23, A105 ASTM A182-23, F6a, F316/F316L ASTM A216-21, WCB ASTM A217-22, WC6, CA15 ASTM A352-21, LCB ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 ASTM A262-15 ASTM G28-22 AGES-GL-13-002,Rev.1 AGES-SP-13-002,Rev.01 1000482-ADD-SP-07-003,Rev.C 1000482-ADD-SP-09-003,Rev.0 NACE MR0175/ISO 15156-2015 Approved GAD	Review the parts Material Test Certificates (In material receiving inspection process)	ASTM A105-23, A105 ASTM A182-23, F6a, F316/F316L ASTM A216-21, WCB ASTM A217-22, WC6, CA15 ASTM A352-21, LCB ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 AGES-GL-13-002,Rev.1 AGES-SP-13-002,Rev.01 1000482-ADD-SP-07-003,Rev.C 1000482-ADD-SP-09-003,Rev.0 NACE MR0175/ISO 15156-2015 Approved GAD Purchase Order CS shall be normalized & SS shall be	Material Test Certification in manufacturer format EN10204-3.2, 3.1 Heat treatment chart, Hardness test report	H	H	W	W1	RW (First lot)

How will it be identified which valve requires 3.1 & 3.2

For valves in 3.2 following steps shall be witnessed by 3.2 TPI
1)Pouring of the cast materials
2)Sample selection
3)Heat treatment
4)Chemical, Mechanical testing -Tensile,impact , hardness testing ,IGC,PMI

Inspection and Test Plan

INSPECTION AND TEST PLAN

Inspection Class: CRITICALITY Ra

Mn-1.35

CEV for butt welded valves shall be 0.42%

MANUFACTURER NAME : SNY VALVE (YANCHENG) CO.,LTD.	**PROJECT / ORDER NO :** EPC WORKS MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM) 1061008020	**ITP NO. :** 1001068020-ITP-R0
DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)	**SUB-SUPPLIER :**	**SUB-ORDER NO. :**

AR Approval Required
A1-A2 Actual Inspection
W1-W2 Witness Inspection
S Operation Surveillance
R1-R2 Review of Documentation
T/C Test Certificate Required
H Hold Point
M Monitoring
1=100%, 2=10%

ACTION	QUALITY RELATED ACTIVITY	REFERENCE DOCUMENT	CHARACTERISTICS TO BE VERIFIED	ACCEPTANCE CRITERIA	VERIFYING DOCUMENT (*Note 1/2)	INTERVENTION POINTS				
						SUB-CONT.	SNY	TPI ADNOC	CONTRACTOR / QC	COMPANY
		Purchase Order		solution annealed & quenched. 1) For A105/WCB/LCB, C≤0.21%, S≤ 0.02%, P≤0.25%,CE ≤0.43. 2) For SS material, C≤0.030% 3) For LCB, impact test results at -46℃, average ≥27 J, individual ≥21J, only one result may be lower than 27 J. 4) Hardness, A105≤187 HBW, WCB & LCB ≤ 197 HBW						
2.3	Reviewing Material Test Certificates of Bonnet Bolt/Nut, Eyebolt pin/Eye bolt/Nut	ASTM A193-23, B7, B7M, B16 ASTM A320-22a, L7, L7M ASTM A194-23, 2H, 2HM, 7, 7M DGS 0000 003 Rev.2 AGES-GL-13-002,Rev.1 AGES-SP-13-002,Rev.01 1000482-ADD-SP-07-003,Rev.C 1000482-ADD-SP-09-003,Rev.0 NACE MR0175/ISO 15156-2015 Approved GAD Purchase Order	Review MTC, check dimensions, PMI, Visual inspection (In material receiving inspection process)	ASTM A193-23, B7, B7M, B16 ASTM A320-22a, L7, L7M ASTM A194-23, 2H, 2HM, 7, 7M DGS 0000 003 Rev.2 AGES-GL-13-002,Rev.1 AGES-SP-13-002,Rev.01 1000482-ADD-SP-07-003,Rev.C 1000482-ADD-SP-09-003,Rev.0 NACE MR0175/ISO 15156-2015 Approved GAD Purchase Order	Material Test Certification in manufacturer format EN10204-3.1	H	H	W1	R1	R1
2.4	Reviewing Material Test Certificates of Packing & Gasket	DGS 0000 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 Approved GAD Purchase Order	Review MTC, check dimensions, PMI, Visual inspection (In material receiving inspection process)	DGS 0000 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 Approved GAD Purchase Order	Material Test Certification in manufacturer format EN10204-2.2	H	H	R		
2.5	Reviewing Material Test Certificates of Welding Consumables (Batch certificate)	DGS 0000 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 Approved GAD Purchase Order	Review MTC, check dimensions, PMI, Visual inspection (In material receiving inspection process)	DGS 0000 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 Approved GAD Purchase Order	Material Test Certification in manufacturer format EN10204-3.1	H	H	R		
3.0	BEFORE MANUFACTURING: MATERIAL RECEIVING INSPECTION									
3.1	Visual Inspection of parts	MSS SP-55 SNY procedure SP-EN3001 Approved GAD Purchase Order	Visual Inspection (In material receiving inspection process)	MSS SP-55 SNY procedure SP-EN3001 Approved GAD Purchase Order	Material Test Certification in manufacturer format EN10204-3.1	H	H	W1	W	R
3.2	Dimensional Inspection of parts	SNY procedure SP-EN3001 & EN3004 Approved GAD Purchase Order	Dimensional Inspection (In material receiving inspection process)	SNY procedure SP-EN3001 & EN3004 Approved GAD Purchase Order	Material Test Certification in manufacturer format EN10204-3.1	H	H	W2	R	R
3.3	Positive Material Identification for alloy steel , stainless steel	DGS 0000 001 Rev.2 ASTM A182-23, F6a, F316/F316L ASTM A217-22, WC6, CA15 ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 AGES-GL-13-002,Rev.1	Positive Material Identification for Alloy steel castings, SS (In material receiving inspection process)	DGS 0000 001 Rev.2 ASTM A182-23, F6a, F316/F316L ASTM A217-22, WC6, CA15 ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 AGES-GL-13-002,Rev.1	PMI Record in manufacturer format	H	H	W2	W	R2

Address corrosion testing requirements

External fasteners (bolts, studs and nuts) shall be coated with PTFE or equivalent. For the details of approved coatings, refer to 1000482-ADD-SP-07-004.

Inspection and Test Plan

INSPECTION AND TEST PLAN

Inspection Class: CRITICALITY Rating:CR1&CR2

MANUFACTURER NAME : SNY VALVE (YANCHENG) CO.,LTD.	**PROJECT / ORDER NO :** EPC WORKS MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM) 1061008020	**ITP NO. :** 1001068020-ITP-R0
DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)	**SUB-SUPPLIER :**	**SUB-ORDER NO. :**

AR	Approval Required
A1-A2	Actual Inspection
W1-W2	Witness Inspection
S	Operation Surveillance
R1-R2	Review of Documentation
T/C	Test Certificate Required
H	Hold Point
M	Monitoring
1=100%, 2=10%	

ACTION	QUALITY RELATED ACTIVITY	REFERENCE DOCUMENT	CHARACTERISTICS TO BE VERIFIED	ACCEPTANCE CRITERIA	VERIFYING DOCUMENT (*Note 1/2)	INTERVENTION POINTS				
						SUB-CONT.	SNY	TPI ADNOC	CONTRACTOR / QC	COMPANY
		Approved PMI procedure Approved GAD Purchase Order		Approved PMI procedure Approved GAD Purchase Order						
3.4	Hardness Testing	ASTM A105-23, A105 ASTM A182-23, F6a, F316/F316L ASTM A216-21, WCB ASTM A217-22, WC6, CA15 ASTM A352-21, LCB ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 1000482-ADD-SP-09-003,Rev.0 NACE MR0175/ISO 15156-2015 Approved GAD Purchase Order	Check part hardness (In material receiving inspection process)	ASTM A105-23, A105 ASTM A182-23, F6a, F316/F316L ASTM A216-21, WCB ASTM A217-22, WC6, CA15 ASTM A352-21, LCB ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 1000482-ADD-SP-09-003,Rev.0 NACE MR0175/ISO 15156-2015 Approved GAD Purchase Order Hardness: A105≤187 HBW, WCB & LCB ≤ 197 HBW	Material Test Certification in manufacturer format EN10204-3.1	H	H	W2	W	R
3.5	NDE-RT on castings	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASME B16.34-2020 Approved GAD Approved NDE procedures Purchase Order Inspection area: Critical area as per ASMR B16.34 para.8.3.1.1.	RT test Parts - body/bonnet for CR-1 Material- WCB, WC6, LCB, 100%	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASME B16.34-2020, Appendix-I Approved GAD Approved NDE procedures Purchase Order	Manufacturer RT test report	H	H	R1	R1	R1
3.6	NDE-MT	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASTM E709-21 ASME B16.34-2020 Approved GAD Approved NDE procedures Purchase Order Inspection area: All accessible surfaces.	MT test Parts - body/bonnet for CR1 & CR2. Material- WCB, WC6, LCB, 100%	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASME B16.34-2020, Appendix-II Approved GAD Approved NDE procedures Purchase Order	Manufacturer MT test report	H	H	R1	W	R1
3.7	NDE-PT,	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASTM E165-23 ASME B16.34-2020 Approved GAD Approved NDE procedures Purchase Order Inspection area: All accessible surface.	PT test Parts - Wedge/Disc/Stem/Seat Ring for CR1 & CR2. Material – CA15, CF3M, CW6MC, F6a, F316L, N06625 100%	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASME B16.34-2020, Appendix-III Approved GAD Approved NDE procedures Purchase Order	Manufacturer PT test report	H	H	R1	R1	R1

WFMT for NACE valves

RT shall be in line with 1000482-ADD-SP-09-003,Rev.0 -clause-9.1

For NACE valves -ASTM supplementary requirements 4 and 5 shall apply for WFMT and RT

Inspection and Test Plan

INSPECTION AND TEST PLAN

MANUFACTURER NAME : SNY VALVE (YANCHENG) CO.,LTD.	**PROJECT / ORDER NO :** EPC WORKS MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM) 1061008020	**ITP NO. :** 1001068020-ITP-R0
DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)	**SUB-SUPPLIER :**	**SUB-ORDER NO. :**

Certain valves in this order has pup pipes at the valve ends . Vendor to details the material of pup pieces as per AGES-SP-09-003

For soft-seated valves, the pup pipes shall be welded to the valve before the valve internals are installed.

The minimum pup piece length shall be as per Attachment -3 of 1000482-ADD-1300-185

Welded ends valves shall be tested along with pup pieces

Vendor shall provide test rings (500mm long) from pup piece Carbon content of butt-welding CS bodies shall be limited to 0.25 % max. The CEV by check analysis shall not exceed 0.42 %.material for field weld procedure qualification in case pup piece is of forged execution.
RT and MPI shall be done for the welding between pup pipie and valve
,WPS/PQR shall be submitted for review and approval
PWHT applicability as per WPS/PQR

Specify where the welding is involved in cast valves

If weld overlay is applicable identify it along with the testings

ACTION	QUALITY RELATED ACTIVITY	REFERENCE DOCUMENT	CHARACTERISTICS TO BE VERIFIED	ACCEPTANCE CRITERIA					
4.0	**DURING MANUFACTURING – WELDING AND ASSEMBLY**								
4.1	Visual Inspection of finished machining parts	SNY procedure SP-EN3002, EN3004 SNY parts Drawings SNY process sheet Approved GAD	Visual Inspection after machining (In-process inspection)	SNY procedure SP-EN3002, EN3004 SNY Drawings SNY process sheet Approved GAD	in manufacturer format	/	H	R1	**R**	**R**
4.2	Welding control- Visual inspection	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 Approved GAD Approved Welding procedures, WPS/PQR Purchase Order	Visual inspection of welding	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 Approved GAD Approved Welding procedures, WPS Purchase Order Major weld repair is not acceptable. Minor weld repair shall be as per WPS/PQR/NDE, PWHT is required.	Welding Inspection records in manufacturer format	/	H	R1	**R**	**R**
4.3	Welding control- Dimensional Inspection & Hardness Testing	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 Approved GAD Approved Welding procedures, WPS Purchase Order	Hardness Testing & Dimensional Inspection on wedge/disc hard facing overlay welding (In-process inspection)	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 Approved GAD Approved Welding procedures, WPS Purchase Order	Welding Inspection records in manufacturer format	/	H	W2	R1	**R**
4.4	Dimensional Inspection of finished machining parts	SNY procedure SNY parts Drawings SNY process sheet Approved GAD	Dimensional Inspection after machining (In-process inspection)	SNY procedure SNY parts Drawings SNY process sheet Approved GAD	Inspection records in manufacturer format	/	H	R1	R1	**R**
4.5	NDE-PT	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASTM E165-23 ASME B16.34-2020 Approved GAD Approved NDE procedures Purchase Order Inspection area: All accessible surface.	PT test - For Machined surfaces of CRA welding overlay 100%	DGS 6300 003 Rev.2 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-6300-001,Rev.0 1000482-ADD-SP-07-003,Rev.C ASME B16.34-2020, Appendix-III Approved GAD Approved NDE procedures Purchase Order	Manufacturer PT test report	/	H	R1	R1	R1
5.0	AFTER MANUFACTURING – VALVE PERFORMANCE TESTING									
5.1	Visual Inspection	MSS SP-55 SNY Assemble Instruction Approved GAD Purchase Order	Visual Inspection on Valves Internal clean	MSS SP-55 SNY Assemble Instruction Approved GAD Purchase Order	Visual Inspection Record in manufacturer format	/	H	R1	W	W
5.2	Positive Material Identification	DGS 0000 001 Rev.2 ASTM A182-23, F6a, F316/F316L ASTM A217-22, WC6, CA15 ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 AGES-GL-13-002,Rev.1 Approved PMI procedure	PMI (Positive Material Identification) on finished Valves	DGS 0000 001 Rev.2 ASTM A182-23, F6a, F316/F316L ASTM A217-22, WC6, CA15 ASTM A351-18e1, CF3M ASTM A494-22, CW6MC ASTM B564-22, N06625 AGES-GL-13-002,Rev.1 Approved PMI procedure	PMI Record in manufacturer format	/	H	W2	W	RW

Vendor shall table all the list of valves as per PO with data sheet number , NACE requiements , material certification 3.1/3.2 and type of valves ,MDMT temperature ,cryogenic valves, vacuum valves

FET requirment to be added to this list

INSPECTION AND TEST PLAN and Test Plan

Inspection Class: CRITICALITY Rating:CR1&CR2

MANUFACTURER NAME : SNY VALVE (YANCHENG) CO.,LTD.	**PROJECT / ORDER NO :** EPC WORKS MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM) 1061008020	**ITP NO. :** 1001068020-ITP-R0
DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)	**SUB-SUPPLIER :**	**SUB-ORDER NO. :**

AR Approval Rec
A1-A2 Actual Inspe
W1-W2 Witness Insp
S Operation Su
R1-R2 Review of Documentation
T/C Test Certificate Required
H Hold Point
M Monitoring
1=100%, 2=10%

Vendor to specify testing requirements for gate , globe and check valves seperately

ACTION	QUALITY RELATED ACTIVITY	REFERENCE DOCUMENT	CHARACTERISTICS TO BE VERIFIED	ACCEPTANCE CRITERIA	VERIFYING DOCUMENT (*Note 1/2)	INTERVENTION POINTS				
						SUB-CONT.	SNY	TPI ADNOC	CONTRACTOR / QC	COMPANY
		Approved GAD Purchase Order		Approved GAD Purchase Order						
5.3	Performance Testing	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase order Test pressure as per approved GAD & Datasheet Test Duration as per API 598	Ambient Pressure Test: 1, Backseat testing 2, Hydro Shell Test 3, Hydro High Pressure Seat Test 4, Gas Low Pressure Seal Test 5, High pressure gas closure test as per DS and gate valve for valves in gas service only with sour toxic & lethal application.	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Acceptance as per API 598 & Clause 10.14 of 1000482-ADD-SP-09-03	Pressure Testing Report in manufacturer format	/	H	W1 (Chloride content of test water shall not exceed 30 ppm and test report to be reviewed by TPI.)	W1	W
5.4	Fugitive Emission Testing (if applicable)	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-SP-07-003,Rev.C 1000482-ADD-1300-030 ISO 15848-2 and MESC SPE 77/312-2024 Test fluid: Helium	Fugitive Emission Tightness & Reviewing Fugitive Emission Prototype testing cert. - For gate valve and globe valve	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-SP-07-003,Rev.C 1000482-ADD-1300-030 ISO 15848-2 and MESC SPE 77/312-2024 Acceptable rate: Class BH, except leakage class AH for all valves with lethal services Approved GAD	FE Test Reports FE Prototype testing certificates	/	H	W1	W1	W
5.5	Functional Test	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase Order	Functional / Operation	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase Order	Functional Test Report	/	H	R1	W	W
5.6	Torque Test	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase Order	Functional / Operation - For one valve of each unique design/size/material	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase Order	Torque Test Report	/	H	R1	RW	RW
5.7	Vacuum Application Test	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 Approved Procedure Approved GAD Purchase Order	Functional / Operation - For one valve of each unique design/size/material	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 Approved GAD Purchase Order Valve shall be tested for vacuum level of 100 mbar (1.45 psi), as per DS.	Torque Test Report	/	H	R1	W	W
5.8	Review Fire safe Certificates and Matrix ~~(if required on purchase order)~~	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase Order	Functional / Operation	1000482-ADD-SP-09-003,Rev.0 1000482-ADD-1300-030 API 598 11th ED (2023) Approved Procedure Approved GAD Purchase Order	Fire safe Certificates and Matrix	/	H	R1	R1	R1

Vacum test is required for all the valves required as per valve data sheet any conflicts shall be approved from JV Engineering

Vendor to confirm cryogenic test is applicability as the drawings are still in approval

Add load test for the valve weight greater than 250 kg

Inspection and Test Plan

INSPECTION AND TEST PLAN

Inspection Class: CRITICALITY Rating:CR1&CR2

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DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)	**SUB-SUPPLIER :**	**SUB-ORDER NO. :**

AR	Approval Required
A1-A2	Actual Inspection
W1-W2	Witness Inspection
S	Operation Surveillance
R1-R2	Review of Documentation
T/C	Test Certificate Required
H	Hold Point
M	Monitoring
1=100%, 2=10%	

ACTION	QUALITY RELATED ACTIVITY	REFERENCE DOCUMENT	CHARACTERISTICS TO BE VERIFIED	ACCEPTANCE CRITERIA	VERIFYING DOCUMENT (*Note 1/2)	INTERVENTION POINTS				
						SUB-CONT.	SNY	TPI ADNOC	CONTRACTOR / QC	COMPANY
6.0	AFTER MANUFACTURING – PAINTING, PACKING AND MARKING									
6.1	Surface Preparation, Adhesion test, Salt Contamination test Final Painting Inspection	Purchase Order Approved Painting Procedure 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-SP-07-004,Rev.C	Coating/Lining Checking Visual Inspection - 100% Adhesion - 1 pc per item DFT - 2pcs per item	Purchase Order Approved Painting Procedure 1000482-ADD-SP-09-003,Rev.0 1000482-ADD-SP-07-004,Rev.C Painting: As per AGES-SP-09-004 A16, Top coat color is Light Grey, RAL 7035.	Coating Inspection Record in manufacturer format	/	H	W2	W	RW
6.2	Final Dimensional Inspection	Approved GAD Purchase Order	Final Dimensional inspection on Gate Valves	Approved GAD Purchase Order	Dimensional inspection Report in manufacturer format	/	H	W1	W	W
6.3	Pre-Shipment Inspection	MSS SP-55 Approved GAD Purchase Order	Finial Visual Inspection on Valves	MSS SP-55 Approved GAD Purchase Order	Visual Inspection Record in manufacturer format	/	H	W2	W2	W
6.4	Marking/Tagging	Approved GAD MSS SP-25 Purchase Order	Marking Checking - Nameplate & Tagging	Approved GAD MSS SP-25 Purchase Order	Nameplate Marking Tagging	/	H	W2	W	W
6.5	Preservation inspection	Purchase Order Approved Procedure 1000482-ADD-SP-09-003,Rev.0 DGS-6710-001 Rev 2	Valve preservation	Purchase Order Approved Procedure 1000482-ADD-SP-09-003,Rev.0 DGS-6710-001 Rev 2	Packing checking record	/	H	R	R	R
6.6	Packing Inspection	DGS 6710 001 Rev.2 1000482-ADD-SP-09-003,Rev.0 Purchase Order Approved Procedure	Packing Checking	DGS 6710 001 Rev.2 1000482-ADD-SP-09-003,Rev.0 Purchase Order Approved Procedure	Packing List Packing Checking Record	/	H	R	W	R
7.0	DOCUMENTS -- FINAL SUBMISSION									
7.1	Document review	DGS 0000 003 Rev.2 General Arrangement Drawings Purchase Order Related material specifications	Inspection and Test records review	DGS 0000 003 Rev.2 General Arrangement Drawings Purchase Order Related material specifications	Material Test Certificates	/	H	R1	AR	AR
7.2	Final release	DGS 0000 003 Rev.2 General Arrangement Drawings Purchase Order ITP	Document Review & Issue of Release Note	DGS 0000 003 Rev.2 General Arrangement Drawings Purchase Order ITP	Manufacturer's Data Book Inspection release note	/	H	H	H	AR
7.3	Manufacturing Record Book (MRB)	DGS 0000 003 Rev.2 General Arrangement Drawings Purchase Order Approved MRB	MTC, NDT reports, EN10204-3.2, 3.1 & 2.2	DGS 0000 003 Rev.2 General Arrangement Drawings Purchase Order	MRB, MTC 10204 3.2/3.1/2.2, Test certificates	/	H	H	AR	AR

ADD MTC review for 3.2 /3.1 certification for the valves

Add stage for Issuance of Inspection Release Note

Inspection and Test Plan

INSPECTION AND TEST PLAN — **Inspection Class: CRITICALITY Rating:CR1&CR2**

MANUFACTURER NAME : SNY VALVE (YANCHENG) CO.,LTD.		**PROJECT / ORDER NO :** EPC WORKS MAXIMIZING ETHANE RECOVERY AND MONETIZATION (MERAM) 1061008020		**ITP NO. :** 1001068020-ITP-R0		AR Approval Required A1-A2 Actual Inspection W1-W2 Witness Inspection S Operation Surveillance R1-R2 Review of Documentation T/C Test Certificate Required H Hold Point M Monitoring 1=100%, 2=10%				
DESCRIPTION : VALVES GATE, GLOBE, CHECK (CAST)		**SUB-SUPPLIER :**		**SUB-ORDER NO. :**						
ACTION	**QUALITY RELATED**	**REFERENCE**	**CHARACTERISTICS TO BE VERIFIED**	**ACCEPTANCE CRITERIA**	**VERIFYING DOCUMENT (*Note 1/2)**	**INTERVENTION POINTS**				
						SUB-CONT.	**SNY**	**TPI ADNOC**	**CONTRACTOR / QC**	**COMPANY**
BY				**REV.NO. / DATE** REV.0 / 22-06-24		**CERTIFIED COMPLETE :**			**ENDORSEMENTS :**	

The Criticality Rating is: 1 and the Inspection Class is: 1 in these cases:
A. All services: 600# and above for all sizes, ; 300# 18" and above.
B. Critical Service (Sour/Cryogenic/Amine/Steam): CS/ LTCS ≥ 6" and / or ≥ 150#
C. CRA valves Critical service (Sour /Cryogenic): all sizes
D. CRA Non sour (e g Chemical injection, Utility, etc) for sizes > 4"
E. Valves for TSO, Hot Tap, stoppling: all sizes and ratings
F. Valves in Cyclic Service: all sizes and ratings

The Criticality Rating is: 2 and the Inspection Class is: 2 in other rating & sizes not covered above.

s qualification certificates shall be reviewed by TPIA during inspection.

nts:

EN 10204 Type 3.2 for body / bonnet ,closing member (Disc) for CR-1, and

EN 10204 Type 3.1 for body / bonnet ,closing member (Disc) for 3.1 for CR-2, and

EN 10204 Type 3.1 for all other metallic parts in contact with fluid, bolts & nuts, and

EN 10204 Type 2.2 for all other parts.

3) Criticality Rating as per AGES-SP-13-002.:

SS Tag plate of the valve shall indicate the name of the manufacturer, model no., serial no., size, rating, service, certification data and valve datasheet no.

In addition - One additional name plate shall be supplied loosely together with valve for insulated valves.

The Globe valve shall be shipped with the obturator in closed position.
Globe Valves shall have the glands and spindles shrouded by a wrapping of petroleum tape around the valve yoke.

Flow direction shall be permanent on the valves
Heat number for the castings shall be permanently marked on the valve body.

Valves shall have a position indicator showing open and closed positions

Valves shall have facilities for provision of padlock at site and for locking in the fully open and fully closed positions.